



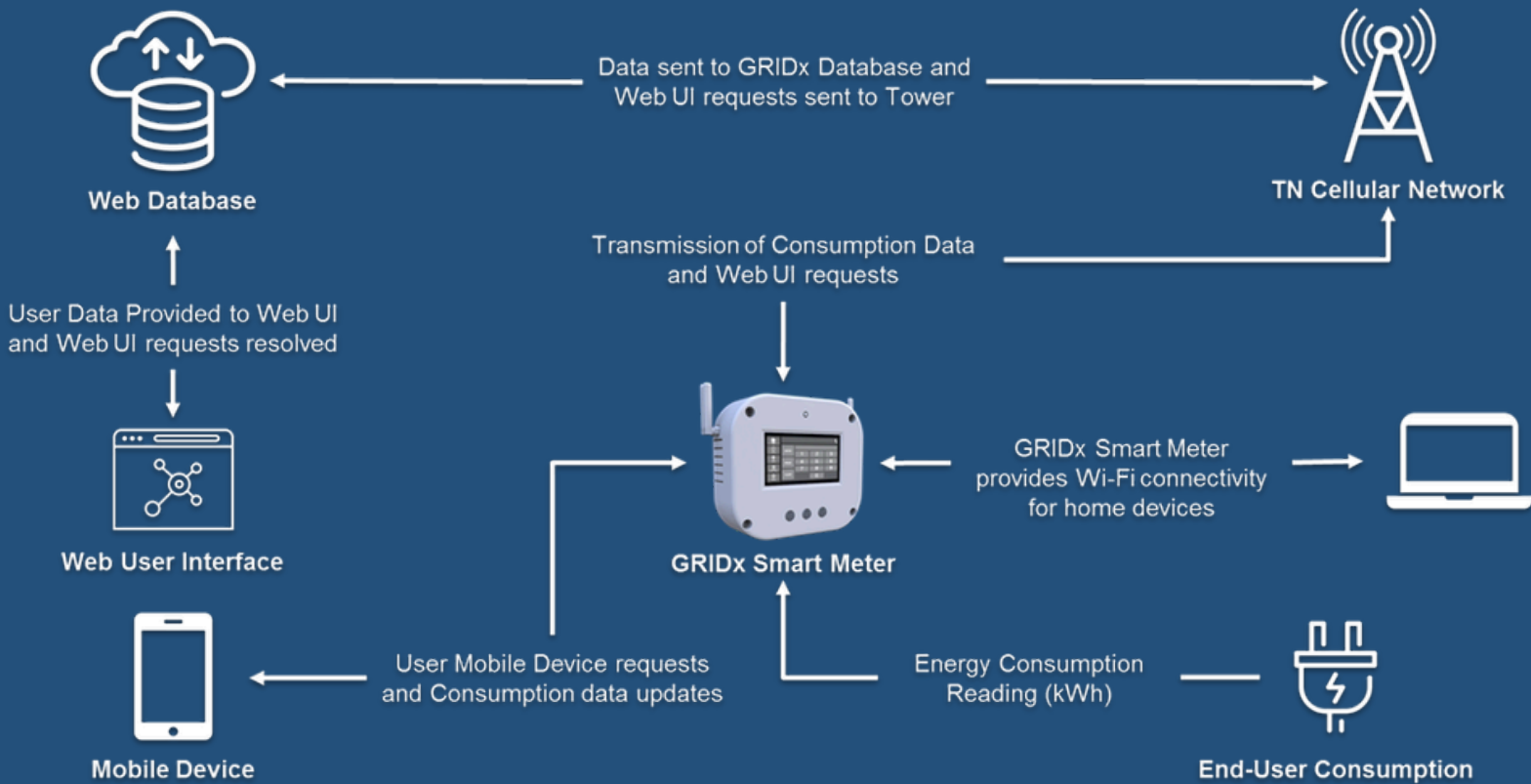
GRIDX

Smart metering platform

One Device. Full Connectivity. Complete Control.

User Manual

Remote Monitoring and Management

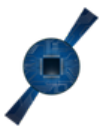


The GRIDx Utility Management Portal provides tools for utilities to remotely monitor and maintain their electric grid.

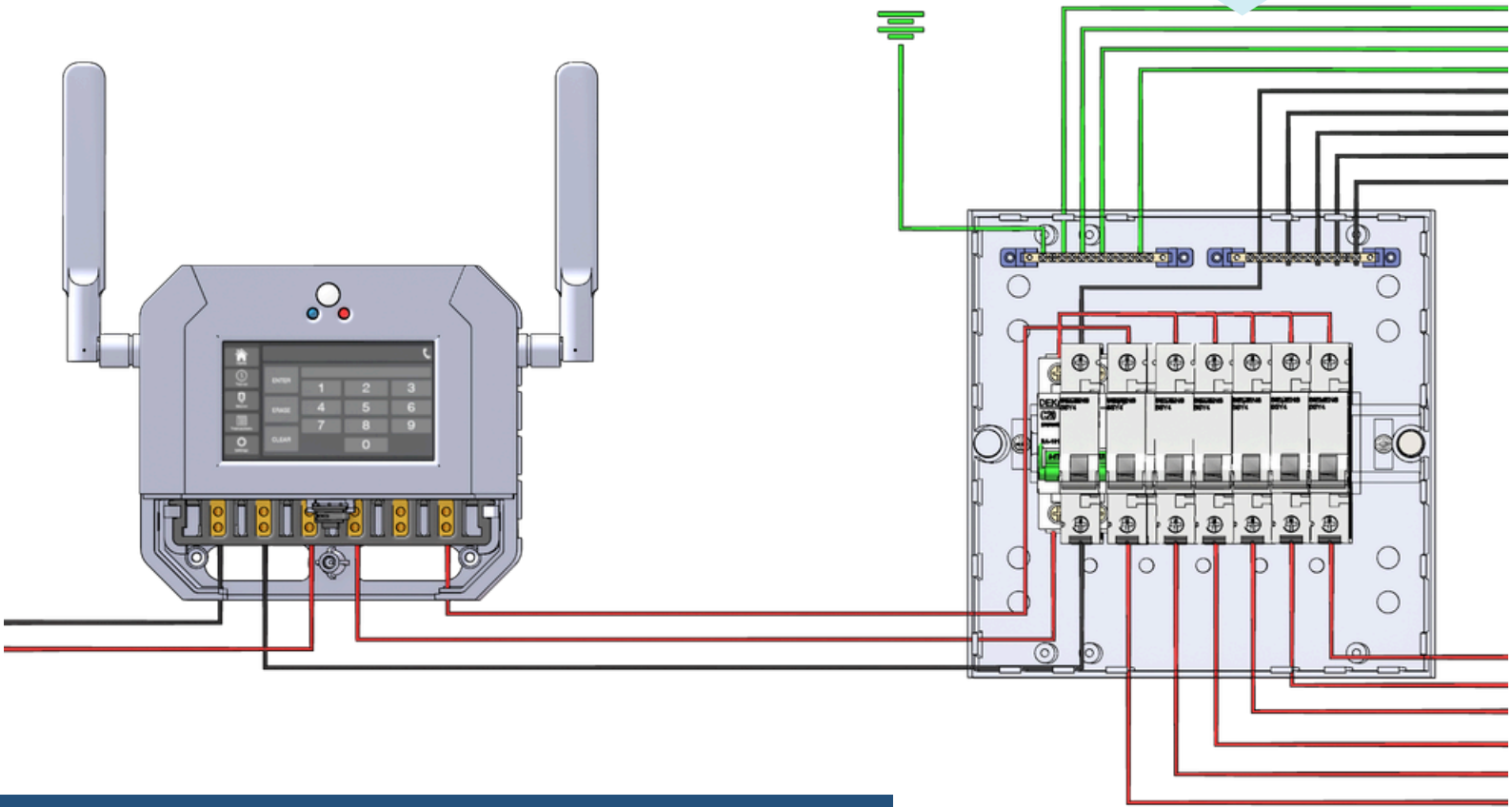
Data generated using an GridX enabled meter gets passed on to an analytics service that processes the data, and allows utilities to run reports on their entire meter population at desired intervals and make informed decisions or take appropriate actions.

The following features of the platform will be monitored in this pilot project:

- Real-time Network Monitoring
- Alerts & Notifications
- Analysis & Reporting
- Meter Management
- Load Analysis
- Grid Topology



Electric Smart Meter

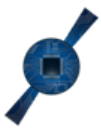


Metering Specifications

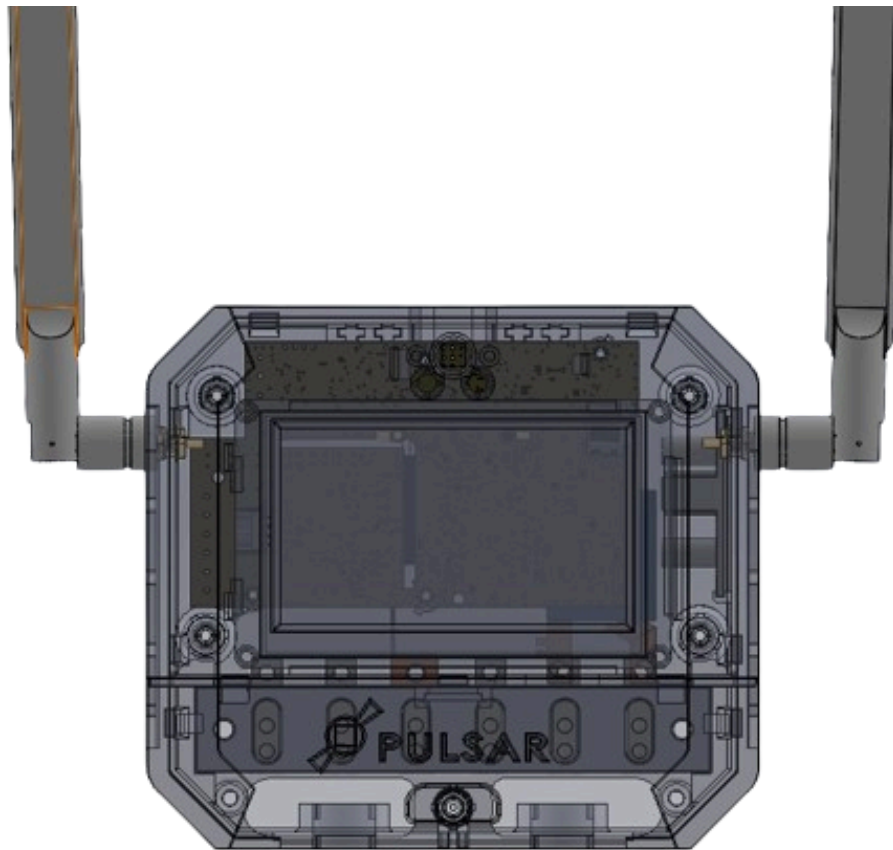
- **Rated Current (A)**
 - 5(80)
- **Rated Voltage (v)**
 - 220 or 240
- **Rated Frequency(Hz)**
 - 50 or 60
- **Accuracy Class**
 - Class 1.0
- **Impulse constant**
 - 1600imp/kWh
- **Temperature**
 - -40 To +125°C
- **Display Type**
 - Touch Screen
 - 3.5 inch
 - 64 MHz
- **Communications**
 - GSM/GPRS+
 - LoRa
 - Bluetooth
- **Load control**
 - Mains control (100Amps)
 - Water heater (60Amps)

Features

- **Autocalibration**
 - Automatic calibration based on a direct measurement of the full signal path
 - Calibration procedure not requiring a reference meter autocalibration Class 1 meter guaranteed
- **3 high performance ADCs**
 - 88 dB SNR
 - 16-bit resolution
- **Advanced metrology feature set**
 - WATT, VAR, VA, Wh, VARh, and VAh
 - Current and voltage rms measurement
 - Power quality measurements
 - Operating temperature, industrial range: -40°C to +115°C
- **Supported international standards**
 - Supports active energy standards: IEC 62053-21
 - IEC 62053-22; EN50470-3;
 - OIML R46; and ANSI C12.20
 - Supports reactive energy standards: IEC 62053-23 and IEC 62053-24
 - Standard transfer specification (STS): IEC 62055-41



Constructions & Materials



Materials

- **Exterior material**
 - ABS's
 - -20 to +80
- **Terminal Block materials**
 - Tufnol
 - +120
 - 1.8 Mpa
- **Dimensions**
 - height: 150mm
 - length: 160mm
 - width : 40mm

Load switching

- **Contacts**
 - **Silver Tin Oxide**(AgSnO₂)
 - Max. switching power
 - 25,000 VA
 - Max. switching
 - voltage 440 VAC
 - Max. switching current
 - 100 A
 - Mechanical life
 - 1,000,000,00 Cycles
- **Insulation & isolation**
 - Creepage and clearance distance
 - 8 mm
 - Test voltage coil - contact
 - 4,000 V
 - Dielectric strength coil
 - 12 kV / 1.2 / 50 μs





Energy Usage Chart



Overview The GRIDx Meter features a dynamic energy usage chart that displays hourly electricity consumption in real-time. This chart helps you track energy usage throughout the day, identify peak usage times, and monitor overall consumption trends.

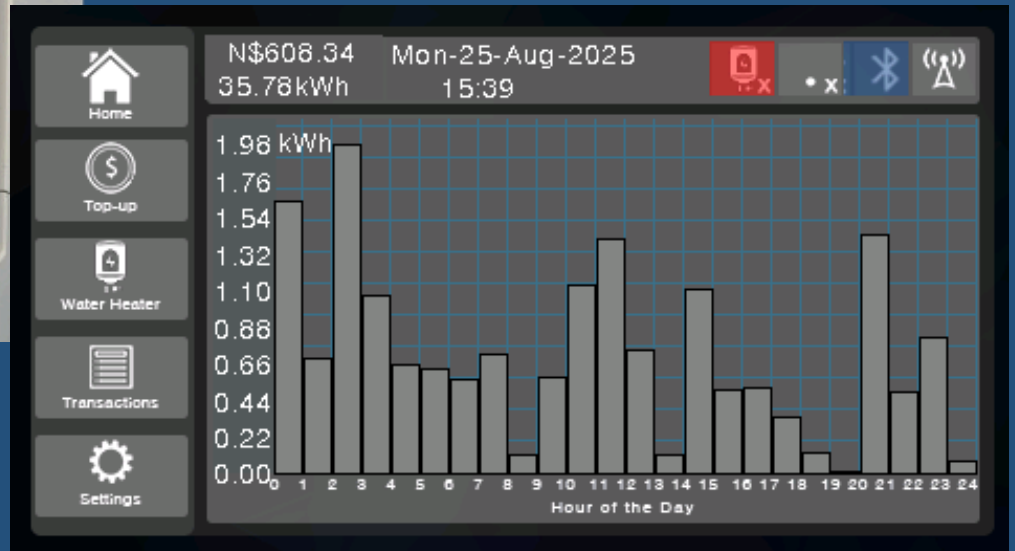


Chart Features

24-Hour Usage Tracking

- Displays energy usage for each hour of the day in watt-hours (Wh), converted to kilowatt-hours (kWh) for readability.

Dynamic Scaling

- The vertical axis (Y-axis) adjusts automatically based on the highest recorded value to ensure clear visualization.

Data Storage

- Hourly consumption values are stored internally and persist across restarts.

Daily Reset

- At midnight, the system clears previous data to start a fresh daily log.

Real-Time Updates

- The chart refreshes approximately every 18 seconds to reflect the latest energy usage data.



Understanding the Display

Bars

- Each bar represents the energy consumed during a specific hour of the day.

Values

- Displayed above each bar in kWh (kilowatt-hours).

Data Storage

- Hourly consumption values are stored internally and persist across restarts.

Y-Axis Markers

- Provide scale references in kWh, dynamically adjusted for visibility.

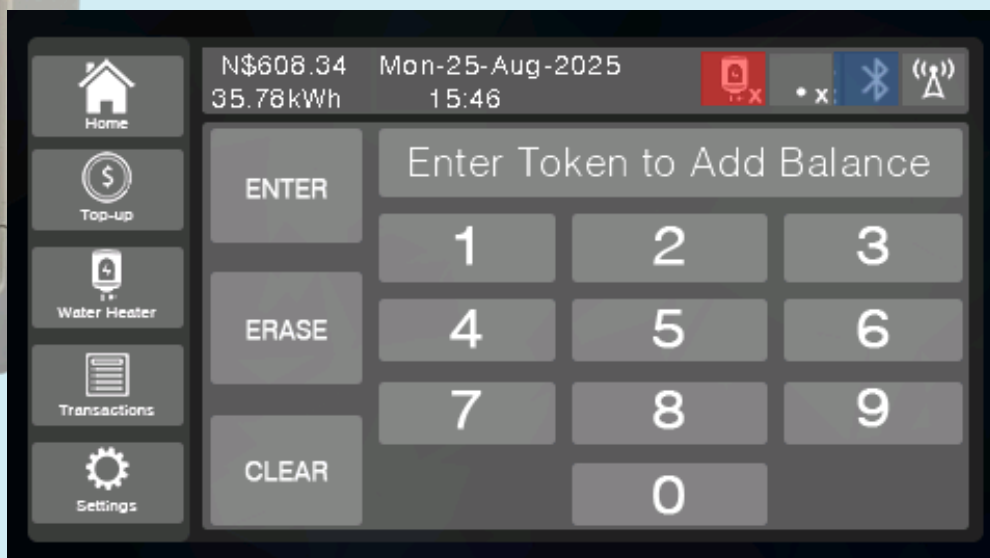
Total Usage & Peak Power

- Logged internally and available for reporting (via telemetry or logs).



Token Processing

The Token Processing Module is responsible for validating, decrypting, authenticating, and executing tokens used in the GRIDx meter. It ensures secure credit management, key changes, proprietary commands, and meter-specific operations.



Keypad Features

Token Authentication

- Validates authenticity using STS standards.

Decryption & Validation

- Ensures token integrity and prevents reuse.

Credit Management

- Adds energy credit via authorized tokens.

Proprietary Commands

- Handles special operations.



System Components

Token Manager

- Handles token storage and management.

TimeKeeping Module

- Logs date and time of token acceptance.

Display & Communication Interfaces:

- Touchscreen Display
- SMS Messaging
- Bluetooth
- API



Token Validation

Accept

- The token has passed all validation checks.

Reject

- The token failed basic validation checks and cannot be processed.

Auth: not authentic

- The token could not be authenticated using the meter's security keys.

Auth: token Used

- The token was valid but has already been processed by this meter.

Validation: token old

- The token is authentic but has a TID (Token Identifier) lower than or outside the acceptable

PROCESSED TOKENS	Mon-25-Aug-2025 15:56
38662336429438366238 01/01/2004, Amount: 245	
35167798725516883867 29/07/2025, Amount: 245	
47291918412576864246 29/07/2025, Amount: 24	
30332669753236821367 30/07/2025, Amount: 1540	
22604761277100952055 30/07/2025, Amount: 54	
<	>
Home	

Geyser Scheduling



The Geyser Scheduling Module allows automated control of a geyser (water heater) based on user-defined schedules, manual operation, or timer mode. It provides an interface via a touchscreen, stores settings in non-volatile memory, and integrates with relay control to switch the geyser ON or OFF at specified times.



Features



Manual Control

- Override automatic scheduling and directly control the geyser.

Daily Scheduling

- Configure geyser ON/OFF times for specific days of the week.

Timer Mode

- Runs geyser for a set countdown period.

Real-Time Monitoring

- Continuously checks time and controls relay accordingly.

Non-Volatile Storage

- Retains geyser settings across power cycles.

Scheduling Configuration



Setting Start and End Times

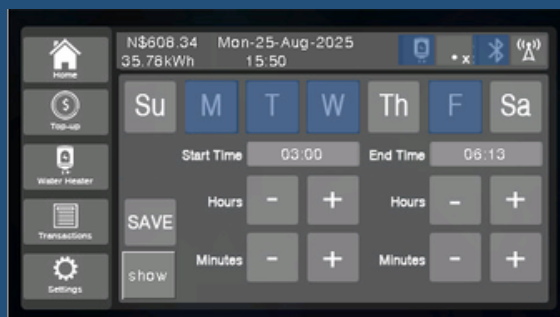
- Hours and minutes are adjusted incrementally using touchscreen buttons.
- Invalid configurations (e.g., start time later than end time) trigger a warning.

Selecting Active Days

- Users can choose which days the geyser schedule should run.
- If no day is selected, the system prompts : "Please select a day"

Saving the Schedule

- Once times and days are set, selecting Save applies changes.
- Feedback is displayed: "Schedule started"
- Schedule applies only to selected days.



Viewing & Deleting Schedules



Viewing Your Schedule

- On your display screen, go to Scheduler.
- You will see a list of all the times your geyser is set to switch ON and OFF.
- Check the days and times to make sure they match your preferences.

Deleting a Schedule

- Go to the Scheduler screen.
- Select the schedule you want to remove.
- Press the Delete button.
- Confirm when asked – the schedule will be removed immediately.



Geyser Timer Control System

Explains how to use the Geyser Timer to set countdown timers, use quick preset times, and control the geyser automatically.

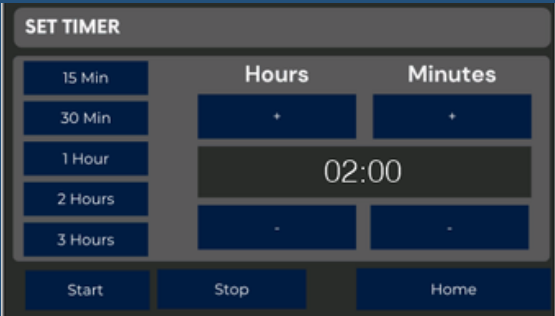


What Does the Timer Do?

- Allows you to set a countdown (e.g., 1 hour 30 minutes).
- Turns the geyser ON when the timer starts.
- Turns the geyser OFF automatically when the timer ends.
- Supports quick timers for common durations (15 min, 30 min, 1 hr, 2 hr, 3 hr).

Timer Display

- Shows the remaining time in HH:MM format.
- Updates every second.
- Resets to 00:00 when the timer stops.



How to Use

Starting the Timer Manually

- Go to the Timer Page on your display.
- Set Hours and Minutes using the + / - buttons.
- Press Start to activate the timer.
- The geyser will turn ON and countdown will begin.

Stopping the Timer

- Press the Stop button anytime to cancel.
- The display resets to 00:00.
- The geyser will turn OFF immediately.

Stopping the Timer

- If connected to a Bluetooth device, the timer value is sent automatically.
- Allows remote monitoring of the remaining geyser time.



This module allows you to add, remove, view, and validate phone numbers authorized to control or receive status updates from the system.

What Does This Feature Do?

- Stores a list of authorized phone numbers in the device's memory
- Ensures only approved numbers can send commands
- Allows for viewing, adding, or removing numbers via the display or remotely.

Adding an Authorized Number

- Input the phone number in the correct format:
 - Local Namibian format: 0812345678 → Stored as +264812345678
 - International format: +264812345678
- The system checks:
 - Minimum of 6 digits.
 - No invalid characters or spaces.
 - Maximum 5 numbers stored.



Viewing Authorized Numbers

- Open the Authorized Numbers page on your display.
- The list shows up to all numbers.
- If no numbers are saved, the screen shows:
 - No authorized numbers saved

Removing a Number

- A specific number - Select it from the list and press Remove.
- All numbers at once - Use the Clear All option (resets the list).

Additional Features

- Some numbers can be marked to receive status updates via SMS.
- These are saved with a [SMS_RESPONSE] tag.

Safety & Limitations

- Only numbers with correct format are saved.
- Maximum 5 authorized numbers.
- Deleting all numbers will remove remote control capability until new numbers are added.



Power Monitoring & Info Display

This module is responsible for reading real-time electrical measurements from the geyser's power monitoring chip and displaying them on the INFO PAGE of the screen.



METER PROFILE	VERSION:0.40.1
Meter Number:	0260066545641
SIM Number:	+264811465041
METER MEASUREMENTS	
Current(A):	0.00
Voltage(V):	0.01
Active(W):	0.00
Reactive(W):	0.00
Temperature:	38.00
Home	

What Does It Do?

- Measures current, voltage, power, energy, temperature, and power factor.
- Updates values on the LCD display in real time.
- Stores data for energy usage reporting and logging.

Live Measurements

Current (A)

- Real-time electrical current.

Voltage (V)

- Line voltage supplied to the meter.

Active Power (W)

- Actual power being used.

Apparent Power (VA)

- Total power including losses.

Temperature (°C)

- Meter internal temperature

System Information

Meter Number (DRN)

- Unique ID of your device.

SIM Number

- GSM SIM number stored in the system.

Software Version

- Current firmware version of the device.

Power Monitoring & Info Display

Your device has three emergency buttons that connect you quickly to help in case of danger, medical issues, or fire.



What Does It Do?

The first press activates a warning message

- Screen shows “Press again to confirm”.
- This prevents false alarms.
- The second press (within a few seconds) confirms the emergency and sends an alert.



Buttons & Their Functions

Police Button

- Press once → Screen says: “Press again to confirm”.
- Press again → A police emergency request is sent.

Ambulance Button

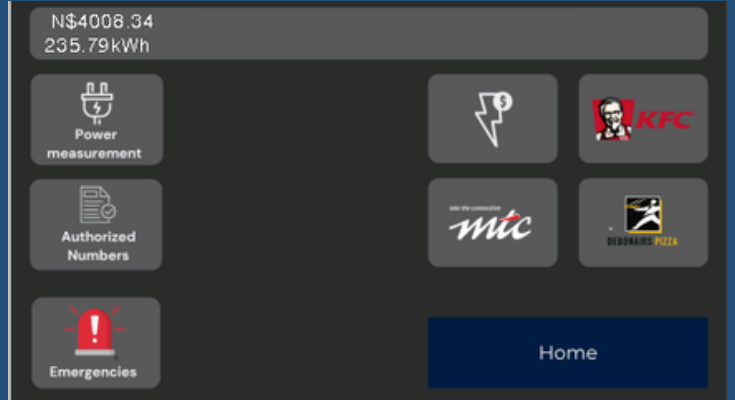
- Press once → Screen says: “Press again to confirm”.
- Press again → An ambulance request is sent.

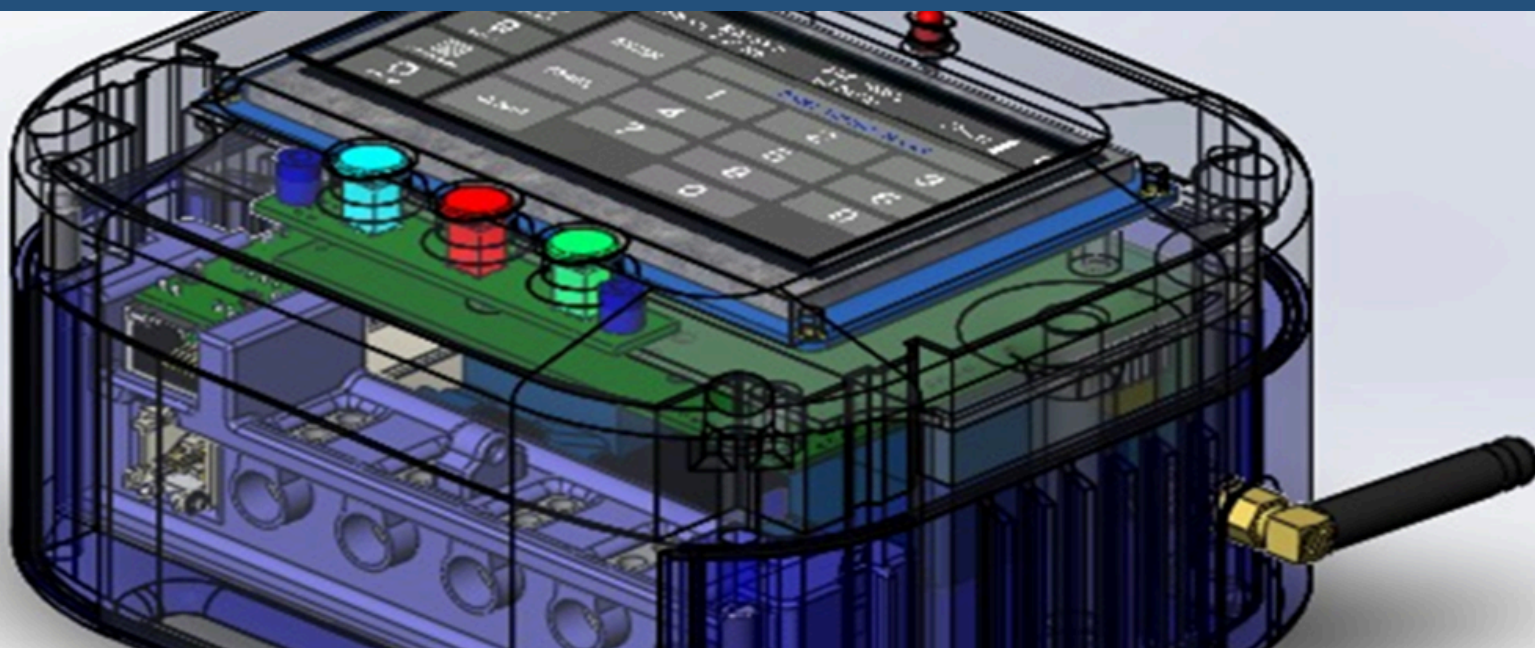
Firefighter Button

- Press once → Screen says: “Press again to confirm”.
- Press again → A firefighter emergency request is sent.

Only press these buttons in real emergencies.

If you press by mistake, do not confirm on the second press – the request will cancel automatically.





Meter Accuracy & Calibration

When meters are installed in residential, municipal and industrial environment they will be subjected to varying conditions over time. These include harsh weather unpredictable loading that will impact the meter.

The conditions will have an impact on the meter's measurement accuracy which may lead to drift in the accuracy of the measured billing quantities resulting in over billing or under billing which could mean lost in revenue.

GRIDx meters provide a solution that consist of a new technology which is integrated in every meter in the field that continuously monitor and reports accuracy of each meter.

In terms of meter accuracy GRIDx provides the utility with the following benefits:

Meter Accuracy

Confirm accuracy of meters in the field and prioritize replacement of meters that are outside of specifications

Meter Accuracy History

Know when drift started and profile progression of problematic meters

Meter Accuracy Prediction

Optimize asset planning through estimating when meters will trend outside specified accuracy limits

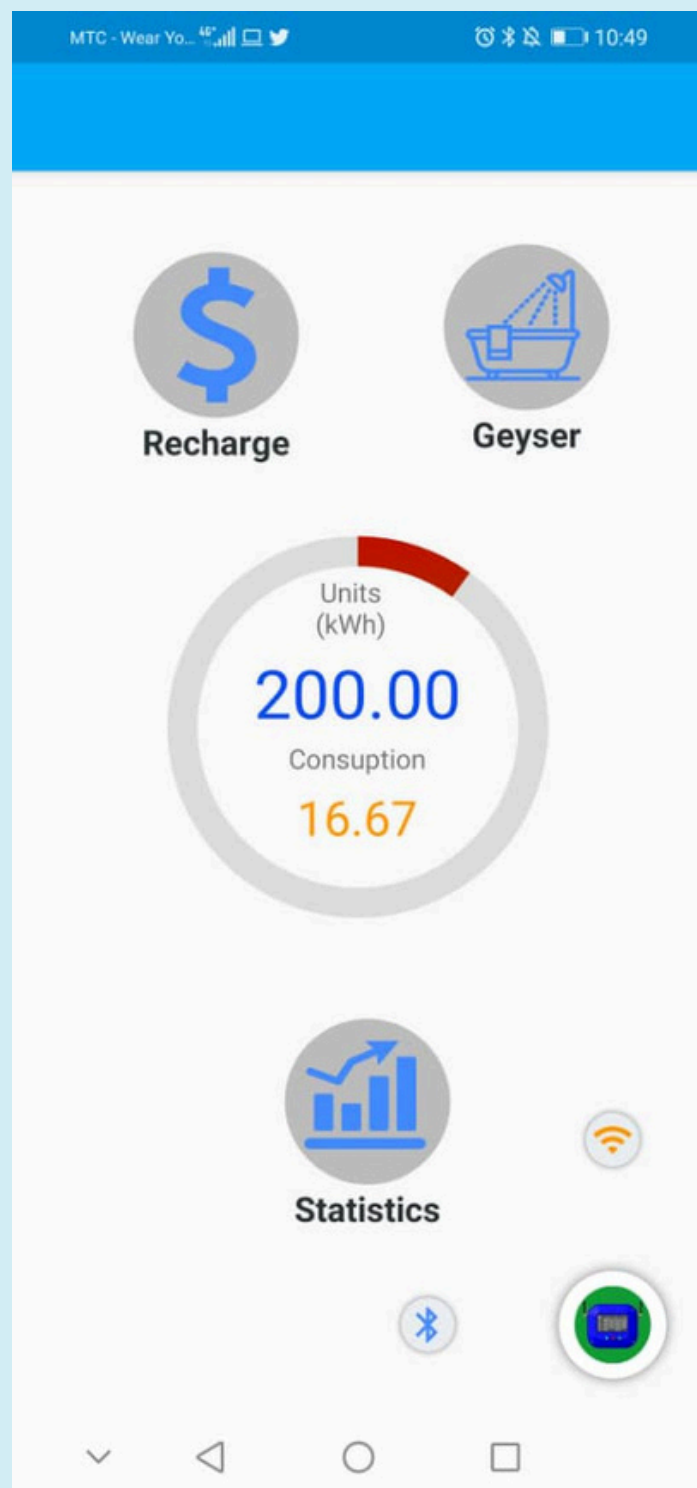
Remote Payment & Load Control

GRIDx meters are equipped with advanced security features to ensure the safe and secure transfer of electricity tokens. The meters support the STS algorithm for remote encryption and decryption of electricity tokens from vending systems. This state-of-the-art encryption method provides an added layer of security to protect against unauthorized access and tampering.

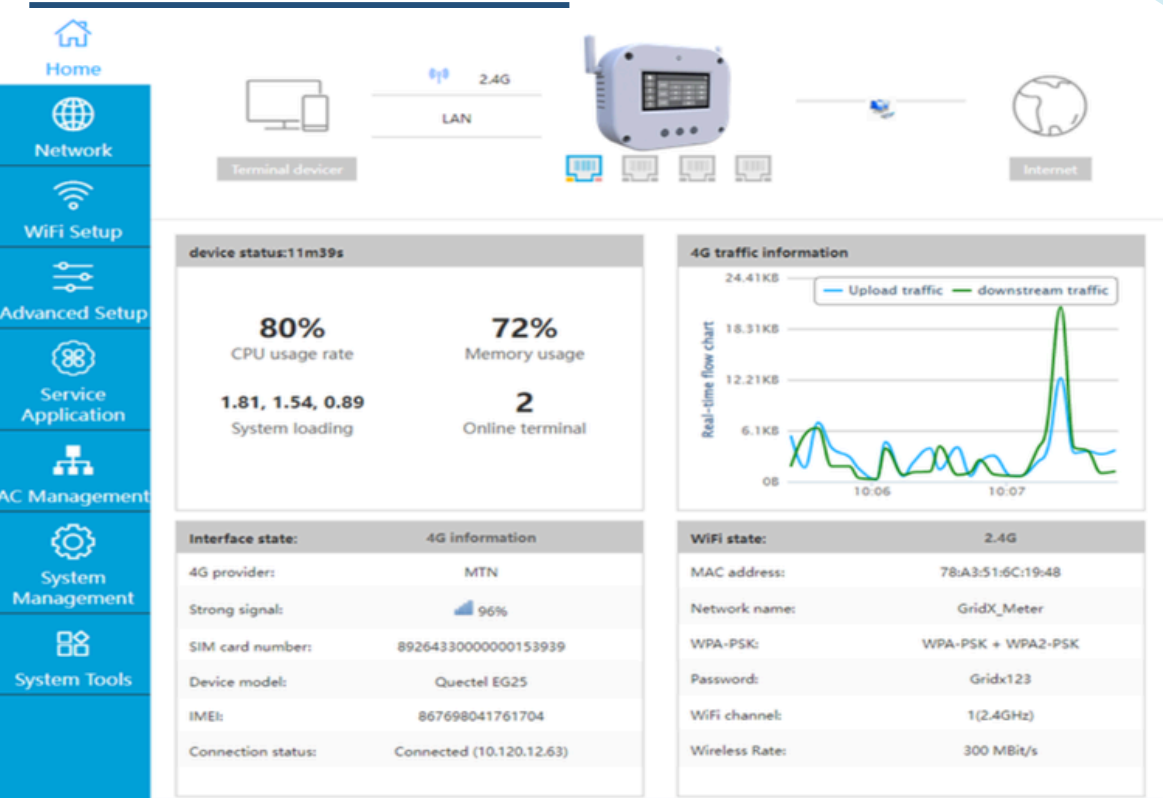
In addition to this, GRIDx meters offer various convenient and flexible methods for remote token transfer, including:

- **Token transfer via SMS, allowing for quick and easy transfer of tokens from anywhere.**
- **Token transfer through a web-based platform, providing a secure and user-friendly interface for administrative purposes.**
- **Token transfer through a Bluetooth channel, enabling seamless transfer of tokens between devices.**
- **Token input through the touch screen, allowing for easy and intuitive token input directly on the meter.**

GRIDX's smart electricity meter not only allows for remote management and control of energy consumption, but also offers the added convenience of remote control of the geyser through a mobile application. This feature allows for load scheduling, enabling users to optimize energy usage and reduce costs by scheduling the operation of their geyser during off-peak hours. This feature is an added benefit that allows more control over your energy consumption and save money on your electricity bill.



Integrated Router



Router Specifications

- **Ram**
 - DDR2 64 MB (max 64MB)
- **Flash**
 - SPI 16 MB (max 32MB)
- **Protocol**
 - IEEE802.11n/802.11g/802.11b/802.3/802.3u
- **Wireless rate**
 - 1Gbps
- **Frequency**
 - 2.4GHZ
 - 5GHZ
- **Antennas**
 - 2 x FPC WIFI antennas, 2*FPC 4G antennas
- **Ports**
 - 1 x 10/100M WAN port support Auto MDI/MDIX;
 - 1 x 10/100M LAN port support Auto MDI/MDIX;
- **POE power supply**

Router Management Interface

Access to the router's management interface is enabled through a LAN (Local Area Network) port, it is required to connect a computer to the router using a network cable. Once the connection is established, it is necessary to configure the computer's network settings to use DHCP for internet access.

The router's management interface can be logged into, where various settings such as network configuration, security, and firmware updates can be accessed. With this feature, network settings can easily be customized and optimal performance can be ensured for the network.